Analyzing architectural microstructure in Transformers through homogenization

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Contents

1 Preliminaries

1.1 Neural Networks

1.1 Definition [Statistical Learning Problem]

Let $(\mathcal{X}, \mathcal{Y})$ be the input and output spaces. Suppose we have a dataset of N independent

samples

$$\mathcal{D} = \Big(x_i, y_i\Big)_{i=1}^{N} \subset \mathcal{X} \times \mathcal{Y},$$

drawn from an unknown joint probability distribution $P_{X,Y}$.
We assume that there exists a true target mapping $(f^\star : x \mapsto y)$ that relates the dataset, up to some possible noise. So, we would like to learn and approximate it by a function parametrized by $\theta \in \Theta$

$$f_\theta : \mathcal{X} \to \mathcal{Y}$$

which minimizes the **expected loss (risk)**

$$\mathcal{R}(\theta) = \mathbb{E}_{(X,Y)\sim P_{X,Y}}\big[r(f_\theta(X), Y)\big],$$

for a fixed elementary loss (risk)

$$r : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_+$$

Since $P_{X,Y}$ is unknown, we typically minimize the **empirical risk**, which we will simply call **loss function**, over the dataset:

$$\widehat{\mathcal{R}}_{\mathcal{D}}(\theta) = \mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N} r\big(f_\theta(x_i), y_i\big)$$

The goal is to find

$$\theta^\star = \arg\min_\theta \mathcal{L}(\theta)$$

And so $f_{\theta^\star}$ would be our approximation of a target mapping.

1.1.1 Feedforward Neural Network

1.2 Definition [Feedforward Neural Network **FFNN**]
Fix depth L. A feedforward neural network consists of layers

$$0, 1, \ldots, L, L+1,$$

where layer 0 is the input layer, layers $1, \ldots, L$ are the hidden layers, and layer $L+1$ is the output layer.

For each layer $\ell = 1, \dots, L+1$, let its dimension be n^ℓ and

$$W^{(\ell)} \in \mathbb{R}^{n_\ell \times n_{\ell-1}}, \qquad b^{(\ell)} \in \mathbb{R}^{n_\ell},$$

be the parameters

$$\theta = \Big((W^{(1)}, b^{(1)}), \dots, (W^{(L+1)}, b^{(L+1)}) \Big),$$

Let $\sigma_\circ^{(\ell)} : \mathbb{R} \to \mathbb{R}$ be an activation function, and $\sigma^{(\ell)} : \mathbb{R}^{n_\ell} \to \mathbb{R}$ its vectorized version. Given an input vector $a^{(0)} \in \mathbb{R}^{n_0}$, the network computes recursively

$$z^{(\ell)} = W^{(\ell)} a^{(\ell-1)} + b^{(\ell)}, \qquad a^{(\ell)} = \sigma^{(\ell)}(z^{(\ell)}), \quad \ell = 1, \dots, L+1.$$

The function realized by the network is the mapping

$$f_\theta : \mathbb{R}^{n_0} \to \mathbb{R}^{n_{L+1}}, \qquad f_\theta(x) = a^{(L+1)} \text{ with } a^{(0)} = x$$

$a^{(0)}$ input vector, $f_\theta(a^{(0)}) = a^{(L+1)}$ output vector / realization,
$z^{(\ell)}$ pre-activation, $a^{(\ell)}$ activation,
$\sigma^{(\ell)}$ vectorized activation function / nonlinearity,
$L+1$ depth, L number of hidden layers, n_ℓ number of neurons at the layer ℓ.

1.3 Example [1 hidden layer FFNN]
Let the depth be $L = 1$. Choose dimensions n_0, n_1, n_2. The network has parameters

$$W^{(1)} \in \mathbb{R}^{n_1 \times n_0}, \qquad b^{(1)} \in \mathbb{R}^{n_1},$$

$$W^{(2)} \in \mathbb{R}^{n_2 \times n_1}, \qquad b^{(2)} \in \mathbb{R}^{n_2},$$

and two activation functions $\sigma^{(1)}$ and $\sigma^{(2)}$, each acting componentwise.
Given an input $x \in \mathbb{R}^{n_0}$, the computation is

$$z^{(1)} = W^{(1)} x + b^{(1)}, \qquad a^{(1)} = \sigma^{(1)}(z^{(1)}),$$

$$z^{(2)} = W^{(2)} a^{(1)} + b^{(2)}, \qquad a^{(2)} = \sigma^{(2)}(z^{(2)}).$$

The function represented by the network is therefore

$$f_\theta(x) = \sigma^{(2)}\big(W^{(2)}\,\sigma^{(1)}(W^{(1)}x + b^{(1)}) + b^{(2)}\big),$$

where $\theta = (W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)})$.

Example of choices: $n_2 = 1,\ \sigma_\circ^{(1)}(t) = t_+,\ \sigma_\circ^{(2)}(t) = (1 + e^{-t})^{-1}$

1.4 Theorem [Cybenko FFNN]

Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a continuous sigmoidal function. Let $f \in C([0,1]^d)$. Then, for any $\varepsilon > 0$ there exist an integer N, real coefficients $a_i, b_i \in \mathbb{R}$, and weight vectors $w_i \in \mathbb{R}^d$ such that

$$\left| f(x) - \sum_{i=1}^{N} a_i\,\sigma\big(w_i \cdot x + b_i\big) \right| < \varepsilon \qquad \forall x \in [0,1]^d$$

In other words, 1 hidden layer FFNN of a sufficient width n_1 can approximate any continuous function.

More general versions of universal approximators theorems exist. They generalize in depth and dimensions, choices for σ, and approximations on the compact subsets of unbounded domains.

1.5 Proposition [Backpropagation]

Consider a feedforward neural network depth L. Let the loss be some regular mapping

$$\mathcal{L}(\theta) = \mathcal{L}\Big(W^{(1)}, b^{(1)}, \ldots, W^{(L+1)}, b^{(L+1)}\Big)$$

with θ denoting all parameters. Let $\sigma^{(\ell)} = \sigma$ and σ' denote the vectorized $\sigma_\circ$. Then,

Output–layer error. *The error at layer $L+1$ is*

$$\delta^{(L+1)} = \bigtriangledown_{a^{(L+1)}}\mathcal{L}\ \odot\ \sigma'\big(z^{(L+1)}\big)$$

Backward error propagation. *For $\ell = L, L-1, \ldots, 1$, the errors satisfy*

$$\delta^{(\ell)} = \Big(W^{(\ell+1)}\Big)^\top \delta^{(\ell+1)}\ \odot\ \sigma'\big(z^{(\ell)}\big)$$

Gradients with respect to parameters. *For each* $\ell = 1, \dots, L+1$,

$$\nabla_{W^{(\ell)}} \mathcal{L} = \delta^{(\ell)} \big(a^{(\ell-1)}\big)^\top \qquad \nabla_{b^{(\ell)}} \mathcal{L} = \delta^{(\ell)}$$

1.6 Proposition [Vanishing Gradient Problem]
Consider a feedforward neural network with hidden depth L. *Let* $\sigma_\circ$ *be differentiable with*

$$0 \le \sigma_\circ'(x) \le c < 1 \quad \text{for all } x \in \mathbb{R}.$$

Then for any $1 \le \ell \le L$,

$$\|\delta^{(\ell)}\| \le c^{L+1-\ell} \left\|W^{(\ell+1)}\right\| \dots \left\|W^{(L+1)}\right\| \|\delta^{(L+1)}\|$$

In particular, if

$$c\,\|W^{(j)}\| < 1 \quad \forall j = \ell+1, \dots, L+1,$$

then

$$\|\delta^{(\ell)}\| \longrightarrow 0 \qquad \text{as } L - \ell \;\to\; \infty,$$

so the gradients with respect to parameters in early layers vanish exponentially with depth.

1.1.2 Other NN architectures

1.7 Definition [Convolutional Neural Network **CNN**]

1.8 Definition [Recurrent Neural Network **RNN**]

1.9 Definition [Residual Neural Network **ResNet**]
Fix depth $L+1$, and for each layer $\ell = 1, \dots, L+1$ denote its dimension n^ℓ and an activation function $\sigma_\circ^{(\ell)}$. Let

$$W^{(\ell)} \in \mathbb{R}^{n_\ell \times n_{\ell-1}}, \qquad b^{(\ell)} \in \mathbb{R}^{n_\ell},$$

be the parameters

$$\theta = \Big((W_1^{(1)}, b_1^{(1)}, W_2^{(1)}, b_2^{(1)}), \dots, (W_1^{(L+1)}, b_1^{(L+1)}, W_2^{(L+1)}, b_2^{(L+1)})\Big),$$

Given an input vector $a^{(0)} \in \mathbb{R}^{n_0}$, the network computes recursively

$$\begin{aligned} z^{(\ell)} &= W_1^{(\ell)} a^{(\ell-1)} + b_1^{(\ell)} \\ a^{(\ell)} &= a^{(\ell-1)} + W_2^{(\ell)} \sigma^{(\ell)}(z^{(\ell)}) + b_2^{(\ell)} \end{aligned}$$

The function realized by the network is the mapping

$$f_\theta : \mathbb{R}^{n_0} \to \mathbb{R}^{n_{L+1}}, \qquad f_\theta(x) = a^{(L+1)} \text{ with } a^{(0)} = x$$

1.10 Definition [Neural ODE for ResNet]
Let $\sigma_\circ$ be an activation function. Let

$$\begin{aligned} F : \mathbb{R}^d \times U &\to \mathbb{R}^d \\ (x, u) &\mapsto W_2\, \sigma(W_1 x + b_1) + b_2 \end{aligned}$$

with control

$$u(t) = (W_1(t), b_1(t), W_2(t), b_2(t)) \in L^\infty([0,1]; U)$$

A neural ODE is then a controlled dynamical system

$$\left| \begin{aligned} \dot{x}(t) &= F(x(t), u(t)) \\ x(0) &= x_0 \end{aligned} \right.$$

So that, the system is driven from an input vector $x(0) \in \mathbb{R}^d$ to a prediction output $x_u(1) \in \mathbb{R}^d$ by u. The training procedure is then the choice of a suitable control u.

1.2 Optimal Transport

1.11 Definition [Monge Problem]

1.12 Definition [Kantorovich Problem]

1.13 Definition [Wasserstein Distances]

1.3 Homogenization

1.3.1 Γ-convergence

1.14 Definition [Γ-convergence]
Let (X, d) be a metric space and let $\varepsilon > 0$ index a family of functionals

$$\mathcal{J}_\varepsilon : X \to \mathbb{R}$$

with a limiting functional

$$\mathcal{J}_0 : X \to \mathbb{R}$$

and the corresponding minimization problems

$$(\mathcal{P}_\varepsilon) : \min_{u \in X} \mathcal{J}_\varepsilon(u) \qquad (\mathcal{P}_0) : \min_{u \in X} \mathcal{J}_0(u)$$

We define a notion of convergence for minimzation problems in the sense that

$$(\mathcal{P}_\varepsilon) \xrightarrow[\varepsilon \to 0]{\Gamma(\mathrm{d})} (\mathcal{P}_0)$$

iff for any $u_0 \in X$ we have

1) $\Gamma - \liminf$, for any sequence $(u_\varepsilon)_\varepsilon$ that converges to u_0 in d, one has

$$\liminf_{\varepsilon \to 0} \mathcal{J}_\varepsilon(u_\varepsilon) \geq \mathcal{J}_0(u_0)$$

2) $\Gamma - \limsup$, there exists a sequence $(u_\varepsilon)_\varepsilon$ that converges to u_0 in d, such that

$$\limsup_{\varepsilon \to 0} \mathcal{J}_\varepsilon(u_\varepsilon) \leq \mathcal{J}_0(u_0)$$

The relaxed version of this definition requires that for any $\eta > 0$, there exists a sequence

$(u_\varepsilon)_\varepsilon$ that converges to u_0 in d, such that

$$\limsup_{\varepsilon\to 0} \mathcal{J}_\varepsilon(u_\varepsilon) \leq \mathcal{J}_0(u_0) + \eta$$

There are even more relaxed versions of this definition, such that the following theorem is still true.

1.15 Theorem
Let (X, d) be a metric space, let

$$(\mathcal{P}_\varepsilon) \xrightarrow[\varepsilon\to 0]{\Gamma(\mathrm{d})} (\mathcal{P}_0)$$

and $(u_\varepsilon)_\varepsilon$ be a family of solutions of $(\mathcal{P}_\varepsilon)_{\varepsilon>0}$ such that

$$u_\varepsilon \xrightarrow[\varepsilon\to 0]{\mathrm{d}} u_0$$

Then, u_0 solves $(\mathcal{P}_0)$.

1.3.2 Periodic Homogenization

1.16 Lemma
Let $b \in L^\infty(\mathbb{R})$ with period 1 and $\phi \in L^1(\mathbb{R})$. Then,

$$\lim_{\varepsilon\to 0} \int_{\mathbb{R}} b\left(\frac{x}{\varepsilon}\right) \phi(x)\,\mathrm{d}x = \langle b\rangle \int_{\mathbb{R}} \phi(x)\,\mathrm{d}x$$

1.17 Example [1-D toy case]

$$\left|\begin{array}{l} -\left[a\left(\dfrac{x}{\varepsilon}\right) u_\varepsilon'(x)\right]' = f(x), \quad x \in (0,1), \\ u_\varepsilon(0) = u_\varepsilon(1) = 0 \end{array}\right.$$

As $\varepsilon \to 0$, it goes to an effective equation

$$\left|\begin{array}{l} -\left[a_\star u_\star'(x)\right]' = f(x), \quad x \in (0,1), \\ u_\star(0) = u_\star(1) = 0 \end{array}\right.$$

with

$$a_\star = \left\langle \frac{1}{a} \right\rangle^{-1} = \left(\int_0^1 \frac{1}{a(x)} \, \mathrm{d}x \right)^{-1}$$

Ansatz

Assume the multiscale expansion of the form

$$u_\varepsilon(x) = u_0(x, x/\varepsilon) + \varepsilon u_1(x, x/\varepsilon) + \varepsilon^2 u_2(x, x/\varepsilon) + \dots$$

After matching coefficients, heuristically gives

$$-\operatorname{\mathbf{div}}_x(A^\star \nabla_x u_0) = f(x)$$

with

$$A^\star_{j,k} = \int_Y \left(A_{j,k}(y) + \sum_{l=1}^{d} A_{j,l} \partial_{y_l} \omega_k(y) \right) \mathrm{d}y$$

where ω_k is a corrector, solving

$$-\operatorname{\mathbf{div}}_y(A(y)\nabla_y \omega_k)(y) = \operatorname{\mathbf{div}}_y(A(y)e_k)$$

Two-scale convergence

1.18 Definition [Two-scale convergence]

1.19 Theorem

1.3.3 Stochastic Homogenization

[Gloria-Otto], [Armstrong-Kuusi-Mourrat]

2 Transformers

[Vas+23]

2.1 ML Perspective

2.1 Definition [Self-Attention Mechanism]
Let $(x_1, \dots, x_n)$ be a list of vectors with $x_i \in \mathbb{R}^d$. Let

$$Q, K, V \in \mathbb{R}^{d \times d}$$

be a triple of linear maps.
Fix a token i and for $j \in [n]$, set the **query** of i, the **key** of j, and the **value** of j

$$q_i = Qx_i, \quad k_j = Kx_j, \quad v_j = Vx_j$$

For a fixed $\beta > 0$, define the similarity kernel

$$w_{ij} = e^{\beta s_{ij}} = e^{\beta \langle q_i \,|\, k_j \rangle}$$

which, after normalization by the partition function,

$$Z_{\beta,i} = \sum_{k=1}^{n} e^{\beta s_{ik}}$$

give the **attention weight** of i on j

$$A_{ij} = \frac{w_{ij}}{Z_i}$$

Finally, compute the new representation token (**context vector**) of i

$$y_i = \sum_{j=1}^{n} A_{ij} v_j$$

Repeat for all $i \in [n]$.

2.2 Definition [Multi-Head Attention]

Let $(x_1, \ldots, x_n)$ be a list of vectors with $x_i \in \mathbb{R}^d$. Let $H \leq d$ and

$$Q_h, K_h, V_h \in \mathbb{R}^{d_h \times d}$$

be a triple of matrices for $h \in [H]$ and $\sum_h d_h = d$.
For each h, compute as before

$$y_i^{(h)} = \sum_{j=1}^{n} A_{ij}^{(h)} v_j^{(h)} \in \mathbb{R}^{d_h}$$

and concatenate

$$y_i = (y_i^{(1)}, \ldots, y_i^{(H)}) \in \mathbb{R}^d$$

Then, for some $O \in \mathbf{SO}(d)$, compute

$$u_j = O(y_j)$$

2.3 Definition [Positional Encoding]
Attention mechanism applied to a prompt $(x_i)_{i \in [n]}$ is permutation invariant. For this, each token is represented as

$$x_i = w_i + p_i$$

Where w is a word embedding and p is a positional encoding that breaks the invariance.

2.4 Definition [Transformer (ML)]

2.2 Dynamical System Perspective

[Ges+25]

2.5 Definition [Transformer (Neural ODE)]
Take a transformer with residual connection. The refined discritization of

$$x^{(\ell+1)} = x^{(\ell)} + y^{(\ell)}$$

will give the dynamics $\dot{x}$, while the RMS normalization will introduce the projector

onto a tangent spaces to a unit sphere
Let $Q, K, V = I_d$

$$\left| \begin{array}{l} \dot{x}_i(t) = \mathsf{P}^{\perp}_{x_i(t)} \left(Z^{-1}_{\beta,i} \sum_{j=1}^{n} e^{\beta \langle x_i(t) \,|\, x_j(t) \rangle} x_j(t) \right) \\ x_i(0) \in \mathbb{S}^{d-1} \end{array} \right. \tag{SA}$$

For general Q, K, V

$$\left| \begin{array}{l} \dot{x}_i(t) = \mathsf{P}^{\perp}_{x_i(t)} \left(\sum_{j=1}^{n} A_{ij}(t) V(t) x_j(t) \right) \\ x_i(0) \in \mathbb{S}^{d-1} \\ A_{ij} = Z^{-1}_{\beta,i} \sum_{k=1}^{n} \exp(\beta \, \langle Q(t) x_i(t) \,|\, K(t) x_k(t) \rangle) \end{array} \right. \tag{SA+}$$

Surrogate model

$$\left| \begin{array}{l} \dot{x}_i(t) = f_i(x) = \mathsf{P}^{\perp}_{x_i(t)} \left(\frac{1}{n} \sum_{j=1}^{n} e^{\beta \langle x_i(t) \,|\, x_j(t) \rangle} x_j(t) \right) \\ x_i(0) \in \mathbb{S}^{d-1} \end{array} \right. \tag{USA}$$

2.6 Definition [Transofmer (Measure-Flow)]

$$\dot{x}_i(t) = \mathcal{V}[\mu(t)](x_i(t))$$

$$\mu(t) = \frac{1}{n} \sum_{i \in [n]} \delta_{x_i(t)}$$

$$Z_{\beta,\mu(t)}(x) = \int e^{\beta \langle x \,|\, y \rangle} \, \mathrm{d}\mu(y)$$

$$\begin{aligned} \mathcal{V}[\mu] : \mathbb{S}^{d-1} &\to \mathsf{T}\mathbb{S}^{d-1} \\ x &\mapsto \mathsf{P}_x \left(Z^{-1}_{\beta,\mu}(x) \int e^{\beta \langle x \,|\, y \rangle} y \, \mathrm{d}\mu(y) \right) \end{aligned}$$

$$\left|\begin{array}{ll} \partial_t \mu + \mathbf{div}(\mathcal{V}[\mu]\mu) = 0, & \mathbb{R}_+ \times \mathbb{S}^{d-1} \\ \mu_{|t=0} = \mu(0), & \mathbb{S}^{d-1} \end{array}\right. \tag{Continuity}$$

2.7 Definition [Interaction Energy]

$$E_\beta[\mu] = \frac{1}{2\beta} \iint e^{\beta\langle x \,|\, x'\rangle} \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(x')$$

3 Transformer Architectures

3.1 Rotary Position Embedding

[Su+23] Instead of

$$x = w + p$$

representation, use rotation

$$x = Rw$$

3.2 Swin Transformer

[Liu+21]

4 Limits

4.1 Number of Particles

$$n \to \infty$$

4.2 Inverse Temperature

$$\beta \to \infty$$

4.3 Spatial Scaling

$$\varepsilon \to 0$$

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